Antian Wang, Ph.D. candidate

Address	The Holcombe Department of Electrical and Computer Engineering Riggs Hall 317, Clemson University Clemson, SC 29634
University E-Mail	antianw@g.clemson.edu
Phone number	678-899-0512
Personal website	`https://wangantian.github.io/`
Research interest	Design automation, Lattice-based cryptography hardware implementation,

Education

2018 – 2024
- **Ph.D. Electrical Engineering (Intelligent System), The Holcombe Department of Electrical and Computer Engineering, College of Engineering, Computing and Applied Sciences, Clemson University SC, U.S.A**
  Advisor: Dr. Yingjie Lao
  Dissertation title: Hardware-oriented Protection and Acceleration for Machine Learning Application.
  Get 30/30 in written format Ph.D. Qualifying Exam in Circuits, Probability and Random Variables, and Signals and Systems.

2018 – 2021
- **Engineering and Science Education Certificate, Department of Engineering and Science Education, College of Engineering, Computing and Applied Sciences, Clemson University SC, U.S.A**
  Eleven Credit Hour courses in Seminar, Pedagogy, Education Research Methods, Professional development, and Practicum.

2013 – 2017
- **B.E. Communication Engineering, Department of Electronic Engineering, College of Information Engineering, Shanghai Maritime University, Shanghai, China.**
  Municipal level Outstanding Graduates
  Outstanding Individual of Construction of Study Style

Research Project

2023 May – current
- **Graduate Research Assistant, The Holcombe Department of Electrical and Computer Engineering, Clemson University.**

Research Publications

Journal Articles

Tan, Weihang, **Antian Wang**, Xinmiao Zhang, Yingjie Lao, and Keshab K. Parhi. "High-Speed VLSI Architectures for Modular Polynomial Multiplication via Fast Filtering and Applications to Lattice-Based Cryptography". In: *IEEE Transactions on Computers* (2023), pp. 1–12.

Tan, Weihang, Benjamin M Case, **Antian Wang**, Shuhong Gao, and Yingjie Lao. "High-Speed Modular Multiplier for Lattice-Based Cryptosystems". In: *IEEE Transactions on Circuits and Systems II: Express Briefs* 68.8 (2021), pp. 2927–2931.

Wang, **Antian**, Weihang Tan, Yuejiang Wen, and Yingjie Lao. "NoPUF: A Novel PUF Design Framework Toward Modeling Attack Resistant PUFs". In: *IEEE Transactions on Circuits and Systems I: Regular Papers* 68.6 (2021), pp. 2508–2521.

Conference Proceedings

Wang, **Antian**, Bingyin Zhao, Weihang Tan, and Yingjie Lao. "NNTesting: Neural Network Fault Attacks Detection Using Gradient-Based Test Vector Generation". In: *Proceedings of the 60th Annual Design Automation Conference 2023*. 2023.

Wang, **Antian**, Weihang Tan, and Yingjie Lao. "Integral Sampler and Polynomial Multiplication Architecture for Lattice-based Cryptography". In: *2022 IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT)*. IEEE. 2022.

Tan, Weihang, **Antian Wang**, Yingjie Lao, Xinmiao Zhang, and Keshab K. Parhi. "Pipelined High-throughput NTT Architecture for Lattice-Based Cryptography". In: *2021 Asian Hardware Oriented Security and Trust Symposium (AsianHOST)*. IEEE. 2021, pp. 1–4.

Tan, Weihang, **Antian Wang**, Keshab K. Parhi, and Yingjie Lao. "Area-Efficient Pipelined VLSI Architecture for Polar Decoder". In: *2020 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*. IEEE. 2020.

Preprint

Tan, Weihang, Sin-Wei Chiu, **Antian Wang**, Yingjie Lao, and Keshab K. Parhi. "PaReNTT: Low-Latency Parallel Residue Number System and NTT-Based Long Polynomial Modular Multiplication for Homomorphic Encryption". 2023. arXiv: 2303.02237.

Patents

Keshab K. Parhi Xinmiao Zhang, Weihang Tan, **Antian Wang**, and Yingjie Lao. *Low Latency Polynomial Modulo Multiplication over Ring*. US Patent 17,582,560, Filed: January 24, 2022. 2022.

Presentation

2022 ISCAS • IEEE International Symposium on Circuits and Systems(ISCAS)
CAS Transactions Paper presentation.

Service

Conference Reviewer

AsianHOST	• IEEE Asian Hardware Oriented Security and Trust Symposium	2021-2023
HOST	• IEEE International Symposium on Hardware Oriented Security and Trust	2023-2024
ISCAS	• IEEE International Symposium on Circuits and Systems	2019-2023
ISVLSI	• IEEE Symposium on VLSI	2022
MWSCAS	• IEEE International Midwest Symposium on Circuits and Systems	2020,2022,2023
SiPS	• IEEE International Workshop on Signal Processing Systems	2019-2020

Journal Reviewer

TCAS-I • IEEE Transactions on Circuits and Systems I: Regular Papers
HaSS • Journal of Hardware and Systems Security
• The Journal of Supercomputing

Teaching

Graduate Teaching Assistant

ECE 2090 • Logic and Computing Devices Laboratory 2022 Fall-2023 Fall.
Perform ABET evaluation material collection and provide suggestions for in-class Lab Kit preparation.
Change the simulation software to the latest Logisim Evolution.
Perform and update all required lab circuit board buildings according to the syllabus.
Enrollment: 71 students for 23 Fall, 35 students for 23 Spring, 41 students for 22 Fall.

ECE 4590/6590 • Integrated Circuit Design 2023 Fall.
Reformat HSPICE lab to using the Cadence SPECTRE tool.
Introduce Cadence RTL to GDSII workflow.
Enrollment: 25 students.

Graduate Grader Assistant

ECE 2010H	• Logic and Computing Devices	2020 Fall.
ECE 2070	• Basic Electrical Engineering	2019 Fall – 2020 Fall.
ECE 2720	• Computer Organization	2020 Fall – 2021 Fall.
ECE 3300	• Signals, Systems, and Transforms	2022 Spring.
ECE 3710	• Micro-controller Interfacing	2020 Fall – 2021 Fall.
ECE 4090	• Introduction to Linear System	2019 Spring, 2019 Fall.
ECE 4270	• Communication Systems	2021 Spring - 2023 Spring.

Miscellaneous Experience

Conference Travel Grant

ISCAS 2022 • ISCAS 2022 Student Participation Grants

Graduate Travel Grant (GTG) from Clemson University Graduate Student Government

2023 Summer	• Design Automation Conference(DAC) 2023, San Francisco, CA.
2022 Summer	• International Symposium on Circuits and Systems(ISCAS) 2022, Austin, TX, USA.
2021 fall	• Design Automation Conference(DAC) 2021, San Francisco, CA/Virtual, USA.
2021 spring	• International Solid-State Circuits Conference(ISSCC) 2021, virtual, USA.

Course taken at Clemson University

CPSC/ECE 6780	• General Purpose Computation on Graphical Processing Units
CPSC 8400	• Design and Analysis of Algorithms
CPSC 8580	• Security in Emerging Computing and Networking Systems
ECE 6590	• Integrated Circuit Design
ECE 6930	• Algorithms for VLSI Design Automation (currently listed as ECE 6580)
ECE 8010	• Analysis of Linear Systems
ECE 8540	• Analysis of Tracking Systems
ECE 8930	• Hardware and AI
ESED 8000	• Seminar in Engineering and Science Education
ESED 8200	• Teaching Undergraduate Engineering

Miscellaneous Experience (continued)

ESED 8500	• Special Topics in Engineering and Science Education (professoriate) (currently offered as ESED 8880 Preparing for the Professoriate)
ESED 8610	• Practicum in Engineering, Science and Mathematics Education
ESED 8720	• Action Research in Engineering, Science and Mathematics Education
MATH 8560	• Theory of Error Correcting Codes
MATH 8570	• Cryptography
MATH 8710	• Machine Learning I
MATH 9850	• Blockchains, Security &Privacy
STAT 8010	• Statistical Methods I
STAT 8030	• Regression and Least Squares Analysis

Skills

Languages	• Strong listening, reading, writing and speaking competencies for English, and Mandarin Chinese.
Coding	• C/C++, Python, R, Matlab, Verilog HDL, VHDL, LaTeX, Tensorflow, PyTorch, SPICE, SPECTRE.